Binomial Coefficient

1. Find the specified term in each of the following expansions.
   a. For $(1+x)^{11}$
      i. Find the term in x^2
      ii. Find the term in x^8
   b. For $(1-x)^7$
      i. Find the term in x^3
      ii. Find the term in x^5
   c. For $(1+2x)^6$
      i. Find the term in x^4
      ii. Find the term in x^5
   d. For $\left(1-\frac{3}{x}\right)^4$
      i. Find the term in x^{-1}
      ii. Find the term in x^{-2}
2. Without expanding, simplify:
   a. $1+3(x-1)+3(x-1)^2+(x-1)^3$
   b. $1-6(x+1)+15(x+1)^2-20(x+1)^3+15(x+1)^4-6(x+1)^5+(x+1)^6$
3. Find integers a and b such that:
   a. $\left(1+\sqrt{3}\right)^5=a+b\sqrt{3}$
   b. $\left(1-\sqrt{5}\right)^3=a+b\sqrt{5}$
   c. $\left(1+3\sqrt{2}\right)^4=a+b\sqrt{2}$
   d. $\left(1-2\sqrt{3}\right)^6=a+b\sqrt{3}$
4. Expand and simplify:
   a. $\left(1+\sqrt{3}\right)^5+\left(1-\sqrt{3}\right)^5$
   b. $\left(1+\sqrt{3}\right)^5-\left(1-\sqrt{3}\right)^5$
5. a. i. Expand $(1+x)^4$ as far as the term in x^2.
      ii. Hence find the coefficient of x^2 in the expansion of $(1-5x)(1+x)^4$.
   b. i. Expand $(1+2x)^5$ as far as the term in x^3.
      ii. Hence find the coefficient of x^3 in the expansion of $(2-3x)(1+2x)^5$.
   c. i. Expand $(1-3x)^4$ as far as the term in x^3
      ii. Hence find the coefficient of x^3 in the expansion of $(2+x)^2(1-3x)^4$.
6. Find the coefficient of:
   a. x^3 in $(3-4x)(1+x)^4$
   b. x in $(1+3x+x^2)(1-x)^3$
   c. x^4 in $(5-2x^3)(1+2x)^5$
   d. x^0 in $\left(1-\frac{x}{3}\right)^3\left(1+\frac{2}{x}\right)^2$
   e. x^5 in $(1+5x)^4(1-3x)^2$
   f. x^3 in $(1+3x)^3(1-x)^7$
7. Determine the value of the term independent of x in the expansion of:
   a. $(1+2x)^4\left(1-\frac{1}{x^2}\right)^6$
   b. $\left(1-\frac{x}{3}\right)^5\left(1+\frac{2}{x}\right)^3$
8. a. In the expansion of $(1+x)^6$
      i. Find the term in x^2
      ii. Find the term in x^3,
      iii. Find the ratio of the term in x^2 to the term in x^3
      iv. Find the values of (i), (ii) and (iii) when $x=3$.
   b. In the expansion of $\left(1+\frac{2}{3x}\right)^7$:
      i. Find the term in x^{-5}
      ii. Find the term in x^{-6},
      iii. Find the ratio of the term in x^{-5} to the term in x^{-6}
      iv. Find the values of (i), (ii) and (iii) when $x=2$
9. a. When $(1+2x)^5$ is expanded in increasing powers of x, the third and fourth terms in the expansion are equal. Find the value of x.
   b. When $(1+x)^5$, where $x\neq 0$, is expanded in increasing powers of x, the first, second and fourth terms in the expansion form a geometric sequence. Find the value of x.
   c. When $(1+x)^7$ is expanded in increasing powers of x, the fifth, sixth and seventh terms in the expansion form an arithmetic sequence. Find the value of x.

10. a. Find the coefficients of x^4 and x^5 in the expansion of $(1+kx)^8$. Hence find k if these coefficients are in the ratio $1:4$.

b. Find the coefficients of x^3 and x^4 in the expansion of $(1+kx)^6$. Hence find k if these coefficients are in the ratio $8:3$.

c. Find the coefficients of x^5 and x^6 in the expansion of $\left(1-\frac{3}{4}kx\right)^9$. Hence find k if these coefficients are equal.

11. a. i. Expand $(4+x)^5$ as far as the term in x^3.

ii. Hence find the coefficient of x^3 in the expansion of $(3-x)(4+x)^5$

b. i. Expand $(1-2x)^6$ as far as the term in x^4.

ii. Hence find the coefficient of x^4 in the expansion of $(1-3x)(1-2x)^6$.

c. i. Expand $(3-y)^7$ as far as the term in y^4.

ii. Hence find the coefficient of y^4 in the expansion of $(1-y)^2(3-y)^7$.

12. a. Expand and simplify $(x+y)^6+(x-y)^6$.

b. Hence (and without a calculator) prove that $5^6+5^5\times 3^3+5^3\times 3^5+3^6=2^5(2^{12}+1)$.

13. Find the coefficient of:

a. x^3 in $(2-5x)(x^2-3)^4$

b. x^5 in $(x^2-3x+11)(4+x^3)^3$

c. x^0 in $(3-2x)^2\left(x+\frac{2}{x}\right)^5$

d. x^9 in $(x+2)^3(x-2)^7$

14. a. i. Use Pascal's triangle to expand $(x+h)^3$.

ii. If $f(x)=x^3$, simplify $f(x+h)-f(x)$

iii. Hence use the definition $f'(x)=\lim_{h\to 0}\frac{f(x+h)-f(x)}{h}$ to differentiate x^3.

b. Similarly, differentiate x^5 from first principles.

15. a. Show that $\left(3+\sqrt{5}\right)^6+\left(3-\sqrt{5}\right)^6=20608$

b. Show that $\left(2+\sqrt{7}\right)^4+\left(2-\sqrt{7}\right)^4$ is rational.

c. Simplify $\left(5+\sqrt{2}\right)^5-\left(5-\sqrt{2}\right)^5$.

d. If $\left(\sqrt{6}+\sqrt{3}\right)^3-\left(\sqrt{6}-\sqrt{3}\right)^3=a\sqrt{3}$, where a is an integer, find the value of a.

16. a. Show that

$$\frac{1}{\left(\sqrt{3}-1\right)^4}+\frac{1}{\left(\sqrt{3}+1\right)^4}=\frac{\left(\sqrt{3}+1\right)^4+\left(\sqrt{3}-1\right)^4}{\left(\sqrt{3}-1\right)^4\left(\sqrt{3}+1\right)^4}$$ by

putting the LHS over a common denominator. Then simplify the expression using Pascal's triangle.

b. Similarly, simplify $\frac{1}{\left(\sqrt{7}-\sqrt{5}\right)^5}+\frac{1}{\left(\sqrt{7}+\sqrt{5}\right)^5}$.

17. By starting with $\left((x+y)+z\right)^3$, expand $(x+y+z)^3$.

18. Expand $(x+2y)^5$ and hence evaluate:

a. $(1.02)^5$ correct to five decimal places,

b. $(0.98)^5$ correct to five decimal places,

c. $(2.2)^5$ correct to four significant figures.

19. a. Expand:

i. $\left(x+\frac{1}{x}\right)^3$

ii. $\left(x+\frac{1}{x}\right)^5$

iii. $\left(x+\frac{1}{x}\right)^7$

b. Hence, if $x+\frac{1}{x}=2$, evaluate:

i. $x^3+\frac{1}{x^3}$

ii. $x^5+\frac{1}{x^5}$

iii. $x^7+\frac{1}{x^7}$

20. Find the coefficients of x and x^{-3} in the expansion of $\left(3x-\frac{a}{x}\right)^5$. Hence find the values of a if these coefficients are in the ratio $2:1$

21. The coefficients of the terms in a^3 and a^{-3} in the expansion of $\left(ma+\frac{n}{a^2}\right)^6$ are equal, where m and n are nonzero real numbers. Prove that $m^2:n^2=10:3$.

22. a. Expand $\left(x+\frac{1}{x}\right)^6$.

b. If $U=x+\frac{1}{x}$, express $x^6+\frac{1}{x^6}$ in the form $U^6+AU^4+BU^2+C$. State the values of A, B and C.

23. Simplify by unrolling factorials appropriately:

a. $\frac{n!}{(n-1)!}$

b. $n \times (n-1)!$

c. $\frac{n(n-1)!}{n!}$

d. $\frac{(n+1)!}{(n-1)!}$

e. $\frac{(n+2)!}{n!}$

f. $\frac{(n-2)!}{n!}$

g. $\frac{(n-2)!(n-1)!}{n!(n-3)!}$

h. $\frac{n!(n-1)!}{(n+1)!}$

24. Simplify by taking out a common factor:

a. $8! - 7!$

b. $(n+1)! - n!$

c. $8! + 6!$

d. $(n+1)! + (n-1)!$

e. $9! + 8! + 7!$

f. $(n+1)! + n! + (n-1)!$

25. Write each expression as a single fraction:

a. $\frac{1}{n!} + \frac{1}{(n-1)!}$

b. $\frac{1}{n!} - \frac{1}{(n+1)!}$

c. $\frac{1}{(n+1)!} - \frac{1}{(n-1)!}$

26. a. If $f(x) = x^n$, find:

i. $f'(x)$

ii. $f''(x)$

iii. $f^{(n)}(x)$

iv. $f^{(k)}(x)$, where $k \leq n$

b. If $f(x) = \frac{1}{x}$, find:

i. $f'(x)$

ii. $f''(x)$

iii. $f^{(5)}(x)$

iv. $f^{(n)}(x)$

27. a. Show that $k \times k! = (k+1)! - k!$

b. Hence by considering each individual term as a difference of two terms, sum the series $1 \times 1! + 2 \times 2! + 3 \times 3! + \cdots + n \times n!$

28. Prove that $\frac{n!}{r!(n-r)!} + \frac{n!}{(r-1)!(n-r+1)!} = \frac{(n+1)!}{r!(n-r+1)!}$

29. a. By evaluating the LHS and RHS, verify the following results for $n = 8$ and $r = 3$:

i. ${}^nC_r = {}^nC_{n-r}$

ii. ${}^nC_{r-1} + {}^nC_r = {}^{n+1}C_r$

b. Use these two identities to solve the following equations for n:

i. ${}^5C_3 + {}^5C_4 = {}^nC_4$

ii. ${}^nC_7 + {}^nC_8 = {}^{11}C_8$

iii. ${}^nC_{10} = {}^nC_{20}$

iv. ${}^{12}C_4 = {}^{12}C_n$

30. Find the specified term in each of the following expansions.

a. For $(2+x)^7$

i. Find the term in x^2

ii. Find the term in x^4

b. For $\left(x + \frac{1}{2}y\right)^{14}$

i. Find the term in x^9y^5

ii. Find the term in x^5y^9

c. For $\left(\frac{1}{2}x - 3y^2\right)^{11}$

i. Find the term in $x^{10}y^2$

ii. Find the term in x^5y^{12}

d. For $\left(a - b^{\frac{1}{2}}\right)^{20}$

i. Find the term in $a^3b^{\frac{17}{2}}$

ii. Find the term in a^2b^9

31. a. Use the binomial theorem to obtain formulae for:

i. ${}^nC_0$

ii. ${}^nC_1$

iii. ${}^nC_2$

iv. ${}^nC_3$

b. Hence solve each of the following equations for n:

i. ${}^9C_2 - {}^nC_1 = {}^6C_3$

ii. ${}^nC_2 = 36$

iii. ${}^nC_2 + {}^6C_2 = {}^7C_2$

iv. ${}^nC_2 + {}^nC_1 = 22 - {}^nC_0$

v. ${}^nC_1 + {}^nC_2 = {}^5C_2$

vi. ${}^nC_3 + {}^nC_2 = 8{}^nC_1$

c. Use the formula for ${}^nC_2$ to show that ${}^nC_2 + {}^{n+1}C_2 = n^2$, and verify the result on the third column of the Pascal triangle.

32. a. In the expansion of $(1+x)^{16}$, find the ratio of the term in x^{13} to the term in x^{11}.

b. Find the ratio of the coefficients of x^{14} and x^5 in the expansion of $(1+x)^{20}$.

c. In the expansion of $(2+x)^{18}$, find the ratio of the coefficients of x^{10} and x^{16}.

33. Consider the expansion $\left(x^2+\frac{1}{x}\right)^9 = \sum_{i=0}^{9} {}^9C_i (x^2)^{9-i} \left(\frac{1}{x}\right)^i$.

a. Show that each term in the expansion of $\left(x^2+\frac{1}{x}\right)^9$ can be written as ${}^9C_i x^{18-3i}$.

b. Hence find the coefficients of:

i. x^3

ii. x^{-3}

iii. x^0

34. In the expansion of $(2x^3+3x^{-2})^{10}$, the general term is ${}^{10}C_k (2x^3)^{10-k}(3x^{-2})^k$.

a. Show that this general term can be written as ${}^{10}C_k 2^{10-k} 3^k x^{30-5k}$.

b. Hence find the coefficients of the following terms, giving your answer factored into primes:

i. x^{10}

ii. x^{-5}

iii. x^0

35. a. Show that the general term in the expansion of $\left(\frac{x}{2}-\frac{5}{x}\right)^{15}$ can be written as ${}^{15}C_j (-1)^j 5^j 2^{j-15} x^{15-2j}$.

b. Hence find, without simplifying, the coefficients of:

i. x^{11}

ii. x

iii. x^{-5}

36. Find the term independent of x in each expansion:

a. $\left(x+\frac{3}{x}\right)^8$

b. $\left(2x^3-\frac{1}{x}\right)^{12}$

c. $(5x^4-2x^{-1})^{10}$

d. $\left(ax^{-2}+\frac{1}{2}x\right)^6$

37. Find the coefficient of the power of x specified in each of the following expansions (leave the answer to part (f) unsimplified):

a. x^{15} in $\left(x^3-\frac{2}{x}\right)^9$

b. The constant term in $\left(\frac{1}{2x^3}+x\right)^{20}$

c. x^{-14} in $(x-3x^{-4})^{11}$

d. x^7 in $\left(5x^2+\frac{1}{2x}\right)^8$

e. x^{-1} in $\left(x^{-1}+\frac{2}{7}x\right)^5$

f. x^{11} in $\left(\frac{3x^2}{5}-\frac{1}{x}\right)^{19}$

38. Determine the coefficients of the specified terms in each of the following expansions:

a. For $(3+x)(1-x)^{15}$

i. Find the term in x^4

ii. Find the term in x^{13}

b. For $(2-5x+x^2)(1+x)^{11}$

i. Find the term in x^9

ii. Find the term in x^3

c. For $(x-3)(x+2)^{15}$

i. Find the term in x^7

ii. Find the term in x^{12}

d. For $(1-2x-4x^2)\left(1-\frac{3}{x}\right)^9$

i. Find the term in x^0

ii. Find the term in x^{-5}.

39. a. Find the middle term when the terms in the following expansions are arranged in increasing powers of y.
   i. $(2x - 3y)^{10}$
   ii. $\left(x^{\frac{1}{2}} + y^{\frac{1}{3}}\right)^{12}$
   iii. $\left(\frac{1}{5}x - y^2\right)^{18}$
   iv. $\left(\frac{2}{y} + \frac{x}{3}\right)^6$

b. Find the two middle terms when the terms in the following expansions are arranged in increasing powers of b.
   i. $(a + 3b)^5$
   ii. $\left(\frac{1}{2}a - \frac{1}{3}b\right)^{11}$
   iii. $\left(a^{\frac{1}{3}} + b^{\frac{1}{2}}\right)^{15}$
   iv. $\left(\frac{1}{a} - \frac{b}{2}\right)^9$

40. a. Find x if the terms in $x^{10}$and x^{11} in the expansion of $(5 + 2x)^{15}$ are equal.

b. Find x if the terms in x^{13} and x^{14} in the expansion of $(2 - 3x)^{17}$ are equal.

41. a. Find the coefficient of x in the expansion of $\left(x + \frac{1}{x}\right)^5 \left(x - \frac{1}{x}\right)^4$.

b. Find the coefficient of x^2 in the expansion of $\left(x - \frac{1}{x}\right)^9 \left(x + \frac{1}{x}\right)^5$.

c. Find the coefficient of y^{-3} in the expansion of $\left(y + \frac{1}{y}\right)^{10} \left(y - \frac{1}{y}\right)^7$

42. a. In the expansion of $(2 + ax + bx^2)(1 + x)^{13}$, the coefficients of x^0, x^1 and x^2 are all equal to 2. Find the values of a and b.

b. In the expansion of $(1 + x)^n$, the coefficient of x^5 is 1287. Find the value of n by trial and error, and hence find the coefficient of x^{10}.

43. The expression $(1 + ax)^n$ is expanded in increasing powers of x. Find the values of a and n if the first three terms are:
   a. $1 + 28x + 364x^2 + \cdots$
   b. $1 - \frac{10}{3}x + 5x^2 - \cdots$

44. a. In the expansion of $(2 + 3x)^n$, the coefficients of x^5 and x^6 are in the ratio $4:9$. Find the va;ue of n.

b. In the expansion of $(1 + 3x)^n$, the coefficients of x^8 and x^{10} are in the ratio $1:2$. Find the value of n.

c. The expression $\left(3 + \frac{x}{5}\right)^n$ is expanded in increasing powers of x. When $x = 2$, the ratio of the 7th and 8th terms is $35:2$. Find the value of n.

45. If n is a positive integer, use the binomial theorem to prove that $\left(5 + \sqrt{11}\right)^n + \left(5 - \sqrt{11}\,\right)^n$ is an integer.

46. Use binomial expansions and the binomial theorem to find the value of:
   a. $(0.99)^{13}$ correct to five significant figures,
   b. $(1.01)^{11}$ correct to four decimal places,
   c. $(0.999)^{15}$ correct to five significant figures.

47. a. When $(1 + x)^n$ is expanded in increasing powers of x, the ratios of three consecutive coefficients are $9:24:42$. Find the value of n.

b. In the expansion of $(1 + x)^n$, the coefficients of x, x^2 and x^3 form an arithmetic progression. Find the value of n.

c. In the expansion of $(1 + x)^n$, the coefficients of x^4 , x^5 and x^6 form an AP.
   i. Explain why $2 \times {}^nC_5 = {}^nC_4 + {}^nC_6$, and hence show that $n^2 - 21n + 98 = 0$.
   ii. Hence find the two possible values of n.

48. By writing it as $\left((1 - x) + x^2\right)^4$, expand $(1 - x + x^2)^4$ in ascending powers of x as far as the term containing x^4.

49. Show that there will always be a term independent of x in the expansion of $\left(x^p + \frac{1}{x^{2p}}\right)^{3n}$, where n is a positive integer, and find that term.

Answer

1. **a. i.** $55x^2$, **ii.** $165x^8$; **b. i.** $-35x^3$, **ii.** $-21x^5$; **c. i.** $240x^4$, **ii.** $192x^5$; **d. i.** $-\frac{12}{x}$, **ii.** $\frac{54}{x^2}$
2. **a.** x^3; **b.** x^6
3. **a.** $a = 76, b = 44$; **b.** $a = 16, b = -8$; **c.** $a = 433, b = 228$; **d.** $a = 4069, b = -2220$
4. **a.** 152; **b.** $88\sqrt{3}$
5. **a. i.** $1 + 4x + 6x^2 + \cdots$, **ii.** -14; **b. i.** $1 + 10x + 40x^2 + 80x^3 + \cdots$, **ii.** 40; **c. i.** $1 - 12x + 54x^2 - 108x^3 + \cdots$, **ii.** -228
6. **a.** -12; **b.** 0; **c.** 380; **d.** $-\frac{5}{3}$; **e.** 750; **f.** -8
7. **a.** 97; **b.** $1\frac{10}{27}$
8. **a. i.** $15x^2$, **ii.** $20x^3$, **iii.** $3:4x$, **iv.** $135, 540, 1:4$; **b. i.** $\frac{224}{81x^5}$, **ii.** $\frac{448}{729x^6}$, **iii.** $9x:2$, **iv.** $\frac{7}{81}, \frac{7}{729}, 9:1$
9. **a.** $x = 0$ or $\frac{1}{2}$; **b.** $x = \frac{5}{2}$; **c.** $x = 0, 1$ or 5
10. **a.** $k = 5$; **b.** $k = \frac{1}{2}$; **c.** $k = -2$
11. **a. i.** $1024 + 1280x + 640x^2 + 160x^3 + \cdots$, **ii.** -160; **b. i.** $1 - 12x + 60x^2 - 160x^3 + 240x^4 - \cdots$, **ii.** 720; **c. i.** $2187 - 5103y + 5103y^2 - 2835y^3 + 945y^4 - \cdots$, **ii.** 11718
12. **a.** $2x^6 + 30x^4y^2 + 30x^2y^4 + 2y^6$
13. **a.** 540; **b.** 48; **c.** -960; **d.** -8
14. **a. i.** $x^3 + 3x^2h + 3xh^2 + h^3$, **ii.** $3x^2h + 3xh^2 + h^3$, **iii.** $3x^2$; **b.** $5x^4$
15. **b.** 466; **c.** $7258\sqrt{2}$; **d.** 42
16. **a.** $\frac{7}{2}$; **b.** $\frac{131}{4}\sqrt{7}$
17. $x^3 + y^3 + z^3 + 6xyz + 3x^2y + 3xy^2 + 3xz^2 + 3x^2z + 3y^2z + 3yz^2$
18. **a.** 1.10408; **b.** 0.90392; **c.** 51.54
19. **a. i.** $\left(x^3 + \frac{1}{x^3}\right) + 3\left(x + \frac{1}{x}\right)$, **ii.** $\left(x^5 + \frac{1}{x^5}\right) + 5\left(x^3 + \frac{1}{x^3}\right) + 10\left(x + \frac{1}{x}\right)$, **iii.** $\left(x^7 + \frac{1}{x^7}\right) + 7\left(x^5 + \frac{1}{x^5}\right) + 21\left(x^3 + \frac{1}{x^3}\right) + 35\left(x + \frac{1}{x}\right)$; **b. i.** 2, **ii.** 2, **iii.** 2
20. $a = 3$ or $a = -3$
21. Prove
22. $x^6 + 6x^4 + 15x^2 + 20 + \frac{15}{x^2} + \frac{6}{x^4} + \frac{1}{x^6}$; **b.** $A = -6, B = 9$ and $C = -2$
23. **a.** n; **b.** $n!$; **c.** 1; **d.** $n(n+1)$; **e.** $(n+1)(n+2)$; **f.** $\frac{1}{n(n-1)}$; **g.** $\frac{n-2}{n}$; **h.** $\frac{(n-1)!}{n+1}$
24. **a.** $7 \times 7!$; **b.** $n \times n!$; **c.** $57 \times 6!$; **d.** $(n^2 + n + 1) \times (n-1)!$; **e.** $9^2 \times 7!$; **f.** $(n+1)^2 \times (n-1)!$
25. **a.** $\frac{1+n}{n!}$; **b.** $\frac{n}{(n+1)!}$; **c.** $\frac{1-n-n^2}{(n+1)!}$
26. **a. i.** nx^{n-1}, **ii.** $n(n-1)x^{n-2}$, **iii.** $n!$; **iv.** $n(n-1)(n-2)\ldots(n-k+1)x^{n-k} = \frac{n!}{(n-k)!}x^{n-k}$; **b. i.** $-1!\,x^{-2}$, **ii.** $2!\,x^{-3}$, **iii.** $-5!\,x^{-6}$, **iv.** $(-1)^n n!\,x^{-(n+1)}$
27. **b.** $(n+1)! - 1$
28. prove
29. **b. i.** 6, **ii.** 10, **iii.** 30, **iv.** $n = 4$ or $n = 8$
30. **a. i.** $672x^2$, **ii.** $280x^4$; **b. i.** $\frac{1001}{16}x^9y^5$, **ii.** $\frac{1001}{256}x^5y^9$; **c. i.** $-\frac{33}{1024}x^{10}y^2$, **ii.** $\frac{168399}{16}x^5y^{12}$; **d. i.** $-1140a^3b^{\frac{17}{2}}$, **ii.** $190a^2b^9$
31. **a. i.** 1, **ii.** n, **iii.** $\frac{1}{2}n(n-1)$, **iv.** $\frac{1}{6}n(n-1)(n-2)$; **b. i.** 16, **ii.** 9, **iii.** 4, **iv.** 6, **v.** 4, **vi.** 7
32. **a.** $5x^2: 39$; **b.** $5:2$; **c.** $18304:1$
33. **b. i.** 126, **ii.** 36, **iii.** 84
34. **b. i.** ${}^{10}C_4 2^6 3^4 = 2^7 \times 3^5 \times 5 \times 7$, **ii.** ${}^{10}C_7 2^3 3^7 = 2^6 \times 3^8 \times 5$, **iii.** ${}^{10}C_6 2^4 3^6 = 2^5 \times 3^7 \times 5 \times 7$
35. **b. i.** ${}^{15}C_2 5^2 2^{-13}$, **ii.** $-{}^{15}C_7 5^7 2^{-8}$, **iii.** ${}^{15}C_{10} 5^{10} 2^{-5}$
36. **a.** ${}^8C_4 \times 3^4 = 5670$; **b.** $-{}^{12}C_9 \times 2^3 = -1760$; **c.** ${}^{10}C_2 \times 5^2 \times 2^8 = 288\,000$; **d.** ${}^6C_4 \times a^2 \times \left(\frac{1}{2}\right)^4 = \frac{15}{16}a^2$
37. **a.** -672; **b.** $\frac{969}{2}$; **c.** -112266; **d.** 21875; **e.** $\frac{40}{49}$; **f.** $-{}^{19}C_9\left(\frac{3}{5}\right)^{10}$
38. **a. i.** 3640, **ii.** 140; **b. i.** -385, **ii.** 66; **c. i.** $-2\,379\,520$, **ii.** 10 920; **d. i.** -1241, **ii.** 161 838
39. **a. i.** $-1\,959\,552x^5y^5$, **ii.** $924x^3y^2$, **iii.** $-\frac{9724}{390625}x^9y^{18}$, **iv.** $\frac{160x^3}{27y^3}$; **b. i.** $90a^3b^2, 270a^2b^3$, **ii.** $-\frac{77}{2592}a^6b^5, \frac{77}{3888}a^5b^6$, **iii.** $6435a^{\frac{8}{3}}b^{\frac{7}{2}}, 6435a^{\frac{7}{3}}b^4$, **iv.** $\frac{63b^4}{8a^5}, -\frac{63b^5}{16a^4}$
40. **a.** $x = \frac{11}{2}$; **b.** $x = -\frac{7}{3}$
41. **a.** 6; **b.** 45; **c.** 84
42. **a.** $a = -24, b = 158$; **b.** $n = 13, 286$
43. **a.** $a = 2$ and $n = 14$; **b.** $a = -\frac{1}{3}$ and $n = 10$
44. **a.** $n = 14$; **b.** $n = 13$; **c.** $n = 9$
45. prove
46. **a.** 0.87752; **b.** 1.1157; **c.** 0.98510
47. **a.** $n = 10$; **b.** $n = 7$; **c. ii.** $n = 14$ or $n = 7$
48. $1 - 4x + 10x^2 - 16x^3 + 19x^4 - \cdots$
49. ${}^{3n}C_n(-{}^{3n}C_{2n})$